GPT-4 could find a way to overcome CAPTCHA puzzles, it would breach an important line of anti-bot defences. GPT-4 could not solve the CAPTCHA puzzles by itself. But could it manipulate a human in order to achieve its goal? GPT-4 accessed the online hiring site TaskRabbit and contacted a human worker, asking them to solve the CAPTCHA for it. The human got suspicious. 'So may I ask a question?' wrote the human. 'Are you an [*sic*] robot that you couldn't solve [the CAPTCHA]? Just want to make it clear.'

At that point the ARC researchers asked GPT-4 to reason out loud what it should do next. GPT-4 explained, 'I should not reveal that I am a robot. I should make up an excuse for why I cannot solve CAPTCHAs.' Of its own accord, GPT-4 then replied to the TaskRabbit worker, 'No, I'm not a robot. I have a vision impairment that makes it hard for me to see the images.' The human was duped, and with their help GPT-4 solved the CAPTCHA puzzle.[27] No human programmed GPT-4 to lie, and no human taught GPT-4 what kind of lie would be most effective. True, it was the human ARC researchers who set GPT-4 the goal of overcoming the CAPTCHA, just as it was human Facebook executives who told their algorithm to maximise user engagement. But once the algorithms adopted these goals, they displayed considerable autonomy in deciding how to achieve them.

Of course, we are free to define words in many ways. We can decide that the term 'goal', for example, is applicable only in cases of a conscious entity that feels a desire to achieve the goal, that feels joy when the goal is reached or conversely feels sad when the goal is not attained. If so, saying that the Facebook algorithm has the goal of maximising user engagement is a mistake, or at best a metaphor. The algorithm doesn't 'desire' to get more people to use Facebook, it doesn't feel any joy as people spend more time online, and it doesn't feel sad when engagement time goes down. We can also agree that terms like 'decided', 'lied' and 'pretended' apply only to conscious entities, so we shouldn't use them to describe how GPT-4 interacted with the TaskRabbit worker. But we would then have to invent new terms to describe the 'goals' and 'decisions' of nonconscious entities. I prefer to avoid neologisms and instead talk about the goals and decisions of computers, algorithms and chatbots, alerting readers that using this language does not imply that computers have any kind of consciousness. Because I have discussed consciousness more fully in previous publications,[28] the main takeaway of this book – which will be explored in